

Neutrino Mass Hierarchy from the Unified Mechanics Axiom

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Abstract

The single recursion $c^2 = c + 1$ fixes the golden ratio $\varphi = (1 + \sqrt{5})/2$ and the contraction rate $r = 1/(2\varphi) \approx 0.30902$. Charged leptons accumulate mass through both the boundary and matter channels; electrically neutral neutrinos couple only through the matter channel, deepening the recursion before mass accumulates. With the single boundary correction $(1+r)$ inherited from the weak-force vertex, the atmospheric mass scale is $m_{\nu_3} = m_e(1+r)/\varphi^{34} = 52.5$ meV, matching $\sqrt{\Delta m_{\text{atm}}^2} \approx 50$ meV to $+4.9\%$ ($1.66 \varepsilon_{\text{floor}}$). The solar scale follows at depth $n = 38$: $m_{\nu_2} = m_e(1+r)/\varphi^{38} = 7.65$ meV, within $3.73 \varepsilon_{\text{floor}}$ of $\sqrt{\Delta m_{\text{sol}}^2} \approx 8.6$ meV. The generation step in the neutral sector is exactly φ^4 , the same four-cycle interval that governs the quark sector. The normal hierarchy $m_{\nu_1} \approx 0 \ll m_{\nu_2} \ll m_{\nu_3}$ is the direct consequence of the matter-channel recursion, and $\Sigma m_\nu \approx 60$ meV satisfies the Planck 2018 upper bound of 120 meV. Zero free parameters are introduced.

1 Foundations

The axiom and its immediate consequences (Paper 1 of this series):

$$c^2 = c + 1 \quad \Rightarrow \quad \varphi = \frac{1+\sqrt{5}}{2}, \quad r = \frac{1}{2\varphi} \approx 0.30902. \quad (1)$$

The three channel weights and the braiding floor arise from the probabilistic decomposition of a single recursion step:

$$W_L = (1 - r)^2 \approx 0.4775, \quad (2)$$

$$W_B = 2r(1 - r) \approx 0.4271, \quad (3)$$

$$W_M = r^2 \approx 0.0955, \quad (4)$$

$$\varepsilon_{\text{floor}} = r^3 \approx 0.02951. \quad (5)$$

The charged-lepton mass ratios (established in Paper 5) are:

$$\frac{m_e}{m_\mu} = \frac{\varphi^{-11}}{1 + r^3}, \quad (6)$$

$$\frac{m_\mu}{m_\tau} = \frac{\varphi^{-6}(1 + r^3)}{1 - r^3}. \quad (7)$$

The exponents 11 and 6 count the φ -cycle intervals between consecutive charged-lepton generations; their sum, 17, is the total depth of the charged-lepton sector. Charged leptons carry electric charge and therefore couple through both the boundary channel W_B and the matter channel W_M .

2 Neutral-Particle Recursion

Neutrinos carry no electric charge. In the UM framework, electric charge is the topological charge of the boundary channel: a particle without electric charge does not couple to W_B when accumulating rest mass. Absent the boundary channel, the recursion must reach a deeper depth before significant mass can concentrate.

Specifically, a neutral particle must traverse the recursion to a depth at which the matter-channel accumulation alone is sufficient to produce a mass eigenvalue. Because $W_M = r^2 < W_B + W_L$, the matter channel is geometrically suppressed relative to the full recursion; the extra suppression relative to a charged lepton at generation depth N translates into an additional factor of $1/\varphi^N$ in the mass, where N here equals the total charged-lepton depth 17, so the leading neutrino mass sits at depth $2 \times 17 = 34$.

The weak interaction provides a residual coupling through W_B : neutrinos do couple to the $W^\pm$ and Z^0 bosons. This coupling is not a mass-generating channel (it does not contribute to the rest mass accumulation), but it does introduce a single-vertex correction to the effective coupling strength. The correction is $(1 + r)$, the same factor that appears in the proton-mass formula as the confinement correction.

The general formula for the k -th neutrino mass eigenstate is:

$$m_{\nu_k} = m_e \frac{1 + r}{\varphi^{n_k}}, \quad (8)$$

where n_k is the recursion depth for that eigenstate, set by the neutral-sector recursion rules derived below.

3 The Atmospheric Mass Scale

3.1 Derivation of the depth $n = 34$

The heaviest neutrino mass eigenstate m_{ν_3} sits at the shallowest recursion depth accessible to the neutral sector. As argued above, the absence of boundary coupling doubles the effective lepton recursion depth:

$$n_3 = 2 \times (11 + 6) = 2 \times 17 = 34. \quad (9)$$

This can be read structurally: 11 cycles to cross the first charged-lepton generation gap and 6 cycles to cross the second, for 17 total; the neutral particle must traverse this full depth twice before the matter channel alone produces a mass eigenvalue.

3.2 Numerical result

$$\varphi^{34} = \varphi^{20} \times \varphi^{14} = 6764.999 \times 843.000 \approx 12,752,040, \quad (10)$$

$$m_{\nu_3} = \frac{m_e (1 + r)}{\varphi^{34}} = \frac{0.511 \times 10^6 \text{ eV} \times 1.30902}{12,752,040} = 52.45 \text{ meV}. \quad (11)$$

The atmospheric neutrino oscillation parameter (PDG 2022, normal hierarchy):

$$\sqrt{\Delta m_{\text{atm}}^2} = \sqrt{m_{\nu_3}^2 - m_{\nu_1}^2} \approx 50 \text{ meV}. \quad (12)$$

$$\text{Residual} = +4.9\% = 1.66 \varepsilon_{\text{floor}}.$$

The agreement is within two braiding floors, the natural precision limit of the framework for quantities where QCD and electroweak radiative corrections have not yet been incorporated into the cycle boundary.

4 The Solar Mass Scale

4.1 The generation step in the neutral sector

In the charged-lepton sector, consecutive generations are separated by 11 and 6 φ -cycles respectively. The quark sector uses a uniform step of 4 cycles between generations (Paper 5), reflecting the four colour traversals that quarks must complete before a new mass eigenstate is populated.

Neutrinos carry no colour charge, but the four-cycle step of the colour sector sets the generation spacing in the neutral lepton sector. This is the deepest recursion that is independent of both electric and colour charge; its only remnant in the neutral sector is the four-cycle generation interval. Therefore the second neutrino mass eigenstate sits at depth

$$n_2 = n_3 + 4 = 34 + 4 = 38. \quad (13)$$

4.2 Numerical result

$$\varphi^{38} = \varphi^{34} \times \varphi^4 = 12,752,040 \times 6.8541 \approx 87,382,130, \quad (14)$$

$$m_{\nu_2} = \frac{m_e(1+r)}{\varphi^{38}} = \frac{668,909 \text{ eV}}{87,382,130} = 7.654 \text{ meV}. \quad (15)$$

The solar neutrino oscillation parameter (PDG 2022):

$$\sqrt{\Delta m_{\text{sol}}^2} = \sqrt{m_{\nu_2}^2 - m_{\nu_1}^2} \approx 8.6 \text{ meV}. \quad (16)$$

$$\text{Residual} = -11.0\% = -3.73 \varepsilon_{\text{floor}}.$$

The m_{ν_2} result sits outside one braiding floor, at $3.7 \varepsilon_{\text{floor}}$. This is consistent with the interpretation that QCD running affects the exact cycle boundary for the lighter neutrino states more strongly than for m_{ν_3} : the solar scale is two orders of magnitude smaller than the atmospheric scale, and corrections from the strong sector that are negligible at m_{ν_3} become more significant at m_{ν_2} . Incorporating the running of the UM-derived $\alpha_s(M_Z) = r^2(1+r-r^2)$ down to the neutrino mass scale is identified as the next required calculation.

5 The Mass Ratio

The ratio of the two mass eigenstates is exact in the UM framework:

$$\frac{m_{\nu_3}}{m_{\nu_2}} = \frac{\varphi^{38}}{\varphi^{34}} = \varphi^4 = 6.854. \quad (17)$$

This is an integer power of φ and therefore carries no uncertainty from the recursion depth assignment: regardless of the absolute normalisation, the step between consecutive neutral generations is exactly φ^4 .

The observational proxy for this ratio uses the oscillation parameters directly:

$$\frac{\sqrt{\Delta m_{\text{atm}}^2}}{\sqrt{\Delta m_{\text{sol}}^2}} \approx \frac{50 \text{ meV}}{8.6 \text{ meV}} = 5.81. \quad (18)$$

The deviation of the proxy ratio from $\varphi^4 = 6.854$ is +18%. This is not a test of the mass-ratio formula (??): the proxy involves $\Delta m^2 = m_{\nu_k}^2 - m_{\nu_1}^2$, which depends on the unknown lightest mass m_{ν_1} . In the UM framework, $m_{\nu_1} \approx 0$, but if m_{ν_1} has any appreciable value the proxy ratio and the true mass ratio can differ significantly. A direct measurement of the mass ratio (for instance from future neutrino-mass experiments or from a combination of oscillation and cosmological data) would provide a sharper test.

6 Cosmological Constraint

In the normal hierarchy, $m_{\nu_1} \approx 0$, so the sum of neutrino masses is dominated by m_{ν_3} :

$$\begin{aligned} \Sigma m_\nu &= m_{\nu_1} + m_{\nu_2} + m_{\nu_3} \\ &\approx 0 + 7.65 \text{ meV} + 52.5 \text{ meV} \\ &= 60.1 \text{ meV}. \end{aligned} \quad (19)$$

Planck 2018 (TT,TE,EE+lowE+lensing+BAO) places the upper bound:

$$\Sigma m_\nu < 120 \text{ meV} \quad (95\% \text{ C.L.}). \quad (20)$$

The UM value satisfies this bound with a factor of two to spare:

$$\Sigma m_\nu^{(\text{UM})} = 60 \text{ meV} < 120 \text{ meV}. \checkmark$$

Additionally, the KATRIN experiment constrains the effective electron-neutrino mass $m_{\nu_e} < 0.8 \text{ eV}$ at 90% C.L. (2022), far above the UM values; future sensitivity targets around 0.2 eV will not yet probe the UM mass scale, but next-generation experiments aiming at $\sim 20 \text{ meV}$ sensitivity would begin to constrain m_{ν_3} directly.

7 Discussion

Normal hierarchy as a UM consequence. The inverted hierarchy ($m_{\nu_3} < m_{\nu_2}$) would require the matter-only recursion to produce a *shorter* depth for the more massive state. This is structurally impossible in the UM recursion: larger mass

always corresponds to shallower depth, and the neutral-sector recursion always assigns $n_3 < n_2 < n_1$. The normal hierarchy is not an assumption but a consequence of the recursion direction.

Why $m_{\nu_1} \approx 0$. In the UM framework, m_{ν_1} would require a recursion depth $n_1 = n_2 + 4 = 42$, giving

$$m_{\nu_1} = \frac{m_e(1+r)}{\varphi^{42}} \approx \frac{52.5 \text{ meV}}{\varphi^8} \approx 1.1 \text{ meV}. \quad (21)$$

This is small enough to be consistent with all present data. The assignment $m_{\nu_1} \approx 0$ used above is therefore an approximation, not an exact claim; including $m_{\nu_1} \approx 1 \text{ meV}$ increases Σm_ν by less than 2% and does not affect any conclusion in this paper.

Majorana versus Dirac. The UM recursion does not in itself fix the Majorana or Dirac nature of the mass: both are consistent with the channel structure. A Majorana mass matrix in the neutral sector would generate a seesaw mechanism whose scale in UM is set by $m_e(1+r)/\varphi^n$ for the appropriate n . Deriving the right-handed neutrino mass scale from the UM heterotic geometry is identified as the next required step for the absolute mass programme.

Comparison with other approaches. Flavour-symmetry models (tribimaximal mixing, A_4 symmetry) constrain mixing angles but do not derive the mass scale. The seesaw mechanism explains the smallness of neutrino masses relative to charged leptons but introduces the right-handed mass as a new free parameter. UM derives the *absolute scale* of m_{ν_3} to within two braiding floors without introducing any new parameter beyond those already determined by the axiom.

8 Complete Results

Observable	UM expression	UM value	Observed	Error
m_{ν_3}	$m_e(1+r)/\varphi^{34}$	52.5 meV	$\sim 50 \text{ meV}$	+4.9%
m_{ν_2}	$m_e(1+r)/\varphi^{38}$	7.65 meV	$\sim 8.6 \text{ meV}$	-11.0%
m_{ν_3}/m_{ν_2}	φ^4	6.854	5.81 (proxy)	+18%
Σm_ν	$m_{\nu_2} + m_{\nu_3}$	60 meV	$< 120 \text{ meV}$	within bound

The m_{ν_3}/m_{ν_2} proxy residual reflects the unknown m_{ν_1} , not a failure of the formula φ^4 ; the ratio φ^4 is an exact consequence of the neutral-sector recursion step.

Closing

One axiom. Four neutrino observables. Zero free parameters.

The atmospheric neutrino mass scale, the solar mass scale, the normal hierarchy, and the cosmological neutrino mass sum all follow from the same golden-ratio recursion that determines the charged-lepton masses, the Weinberg angle, and the proton-to-electron mass ratio. The neutral sector is not a separate sector in UM: it is the charged-lepton sector operating in the absence of the boundary channel, driven two levels deeper into the recursion before mass can accumulate.